



Coffee Shop Sales Analytics Dashboard (Power BI)



Short Description / Purpose

An end-to-end sales analytics dashboard built in Power BI to evaluate the performance of a multi-location Coffee Shop chain across **3 store locations** in New York City. The dashboard enables operations and business teams to monitor revenue trends, identify peak sales periods, understand product performance, and make data-driven decisions to improve profitability and customer experience.



Tech Stack

- **Power BI Desktop** — Dashboard development and report publishing
- **DAX (Data Analysis Expressions)** — KPI measures, MoM calculations, time intelligence
- **Power Query (M Language)** — Data cleaning, type conversion, transformation
- **MySQL** — Data extraction, pre-analysis, and KPI validation queries
- **Excel (.xlsx)** — Source data file
- **Star Schema Data Modelling** — Date_Table + Transactions fact table
- **Data Visualization & KPI Design** — Calendar heatmap, bar charts, donut charts, matrix heatmap



Data Source

Dataset contains **149,116 real transaction records** across January to June 2023 from a coffee shop chain with 3 New York City locations.

Table: Transactions (Fact Table)

Column	Description
transaction_id	Unique transaction identifier
transaction_date	Date of transaction (Jan–Jun 2023)
transaction_time	Time of transaction (hour level)
transaction_qty	Quantity of items sold
store_id	Store identifier
store_location	Hell's Kitchen / Astoria / Lower Manhattan
product_id	Product identifier
unit_price	Price per unit
product_category	Coffee, Tea, Bakery, Drinking Chocolate, etc.
product_type	Barista Espresso, Brewed Chai Tea, Hot Chocolate, etc.
product_detail	Full product name and size

Supporting Table: Date_Table (Dimension) Custom calculated date table built in DAX for time intelligence functions.

Features / Highlights

Business Problem

The coffee shop management team had no unified view of sales performance across their 3 locations. Key questions like — *Which store is performing best? What hours drive the most revenue? Which products should be promoted?* — could not be answered without manual, time-consuming reporting.

Goal of the Dashboard

To build a dynamic, interactive sales performance dashboard that gives management a clear, real-time picture of revenue, orders, and quantity sold — broken down by location, product, time of day, and day of week — enabling faster and smarter operational decisions.

Walkthrough of Key Visuals

Page 1 — Sales Report (Main Dashboard)

Visual	What It Shows
KPI Cards	Total Sales (\$698K), Total Orders (149K), Total Qty Sold (214K) with MoM % change
Month Filter + Calendar	Interactive month selector with a calendar chart showing daily sales intensity
Sales Trend (Bar Chart)	Daily sales trend for selected month vs average sales line
Sales by Weekend vs Weekday (Donut)	Weekend 27.94% vs Weekday 72.06% revenue split
Sales by Store Location (Bar)	Revenue comparison across Hell's Kitchen, Astoria, Lower Manhattan
Sales by Product Category (Bar)	Revenue ranked by category — Coffee leads at \$270K
Top Product Types (Bar)	Top 10 product types by revenue
Sales by Day & Hour (Matrix Heatmap)	Hour vs Day grid showing peak revenue combinations

Page 2 — Tooltip: Calendar Chart Hover-over tooltip showing daily sales, orders, and quantity for any date on the calendar.

Page 3 — Tooltip: Day & Hour Chart Hover-over tooltip showing hour-level breakdown for any day-hour combination in the matrix.

Business Impact & Insights

Overall Performance (Jan–Jun 2023)

Metric	Value
Total Revenue	\$698,812
Total Orders	149,116

Metric	Value
Total Qty Sold	214,470
Average Order Value	\$4.69



Monthly Revenue Trend — Consistent Growth

Month	Revenue	Orders	MoM Growth
January 2023	\$81,678	17,314	—
February 2023	\$76,145	16,359	-6.8%
March 2023	\$98,835	21,229	+29.8%
April 2023	\$118,941	25,335	+20.3%
May 2023	\$156,728	33,527	+31.8%
June 2023	\$166,486	35,352	+6.2%


Insight: Revenue nearly doubled from January to June — a 103% growth over 6 months, indicating strong business momentum. The February dip is likely seasonal; the sharp March rebound suggests a promotional or seasonal upturn.



Sales by Store Location

Location	Revenue	Orders
Hell's Kitchen	\$236,511	50,735
Astoria	\$232,244	50,599
Lower Manhattan	\$230,057	47,782


Insight: All three locations perform remarkably evenly — within 3% of each other — indicating consistent brand standards and customer experience across stores. Hell's Kitchen holds a slight edge in both revenue and footfall.



Sales by Product Category

Category	Revenue	Share
Coffee	\$269,952	38.6%
Tea	\$196,406	28.1%
Bakery	\$82,316	11.8%
Drinking Chocolate	\$72,416	10.4%
Coffee Beans	\$40,085	5.7%

Category	Revenue	Share
Branded	\$13,607	1.9%
Loose Tea	\$11,214	1.6%
Flavours	\$8,409	1.2%
Packaged Chocolate	\$4,408	0.6%

 **Insight:** Coffee and Tea together account for **66.7%** of total revenue. Bakery is a strong third — suggesting bundled promotions (coffee + pastry) could further lift average order value.

Top 10 Product Types by Revenue

Rank	Product Type	Revenue
1	Barista Espresso	\$91,406
2	Brewed Chai Tea	\$77,082
3	Hot Chocolate	\$72,416
4	Gourmet Brewed Coffee	\$70,035
5	Brewed Black Tea	\$47,932
6	Brewed Herbal Tea	\$47,540
7	Premium Brewed Coffee	\$38,781
8	Organic Brewed Coffee	\$37,747
9	Scone	\$36,866
10	Drip Coffee	\$31,984

 **Insight:** Barista Espresso alone contributes **13.1%** of total revenue — making it the single most critical product to stock, staff for, and promote. Brewed Chai Tea's strong second place shows that tea drinkers are a key customer segment not to overlook.

Sales by Hour — Peak Trading Hours

Hour	Revenue	Performance
10:00 AM	\$88,673	 Peak
9:00 AM	\$85,170	 Peak
8:00 AM	\$82,700	 Peak
7:00 AM	\$63,526	 High
11:00 AM	\$46,319	 Moderate
3:00 PM	\$41,733	 Moderate

Hour	Revenue	Performance
2:00 PM	\$41,305	● Moderate
4:00 PM	\$41,123	● Moderate

 **Insight:** The **morning rush (7 AM–10 AM)** drives the highest revenue — accounting for approximately **45% of daily revenue** in just 4 hours. Staffing levels, inventory prep, and queue management should be prioritised for this window. The afternoon (2–4 PM) shows a secondary bump worth targeting with afternoon specials.

Sales by Day of Week

Day	Revenue
Monday	\$101,677
Friday	\$101,373
Thursday	\$100,768
Wednesday	\$100,314
Tuesday	\$99,456
Sunday	\$98,330
Saturday	\$96,894

 **Insight:** Sales are **remarkably consistent across all 7 days** — varying by less than 5% from busiest (Monday) to quietest (Saturday). This is unusual and positive — it means the business is not weekend-dependent. However, **Saturday being the lowest revenue day** is counterintuitive and worth investigating — could reflect location demographics (office crowd in some areas drives weekday traffic).

Weekend vs Weekday Split

Period	Revenue	Orders	Share
Weekdays (Mon–Fri)	\$503,588	107,510	72.06%
Weekends (Sat–Sun)	\$195,225	41,606	27.94%

 **Insight:** Weekdays generate **2.6x more revenue** than weekends — strongly suggesting the customer base is office workers and commuters rather than leisure visitors. Weekend marketing campaigns or family-oriented offers could help close this gap.

Top 5 Individual Products by Revenue

Product	Revenue
Sustainably Grown Organic Lg	\$21,152

Product	Revenue
Dark Chocolate Lg	\$21,006
Latte Rg	\$19,112
Cappuccino Lg	\$17,642
Morning Sunrise Chai Lg	\$17,384

 **Insight:** Large-size premium products dominate the top 5 — customers are willing to pay for quality and size. Promoting upsizing and premium variants through loyalty programmes could meaningfully lift AOV beyond the current \$4.69.

MySQL Queries Used

All data cleaning, type conversion, and KPI validation was performed in MySQL before building the Power BI model. Below are the actual queries used in this project.

Data Preparation & Cleaning

```
-- Create database
CREATE DATABASE coffee_shop_sales_db;

-- Fix BOM character issue in column name
ALTER TABLE coffee_shop_sales
CHANGE COLUMN `ï»¿transaction_id` transaction_id INT;

-- Convert transaction_date string to proper DATE format
UPDATE coffee_shop_sales
SET transaction_date = STR_TO_DATE(transaction_date, '%d-%m-%Y');

ALTER TABLE coffee_shop_sales
MODIFY COLUMN transaction_date DATE;

-- Convert transaction_time string to proper TIME format
UPDATE coffee_shop_sales
SET transaction_time = STR_TO_DATE(transaction_time, '%H:%i:%s');

ALTER TABLE coffee_shop_sales
MODIFY COLUMN transaction_time TIME;

-- Disable safe update mode for bulk updates
SET SQL_SAFE_UPDATES = 0;

-- Verify column data types
DESCRIBE coffee_shop_sales;
```

Q1 — Total Sales KPI + MoM Growth

```
-- Total Sales for May
SELECT CONCAT(ROUND(SUM(unit_price * transaction_qty)) / 1000, 'K') AS Total_Sales
```

```

FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5; -- May Month

-- MoM Sales Difference & Growth % (April vs May)
SELECT
    MONTH(transaction_date) AS Month,
    ROUND(SUM(unit_price * transaction_qty)) AS Total_Sales,
    (SUM(unit_price * transaction_qty) - LAG(SUM(unit_price * transaction_qty), 1)
     OVER (ORDER BY MONTH(transaction_date))) /
    LAG(SUM(unit_price * transaction_qty), 1)
     OVER (ORDER BY MONTH(transaction_date)) * 100 AS MOM_Increase_Percentage
FROM coffee_shop_sales
WHERE MONTH(transaction_date) IN (4, 5) -- April (PM) and May (CM)
GROUP BY MONTH(transaction_date)
ORDER BY MONTH(transaction_date);

```

Q2 — Total Orders KPI + MoM Growth

```

-- Total Orders for May
SELECT COUNT(transaction_id) AS Total_Orders
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5;

-- MoM Orders Difference & Growth %
SELECT
    MONTH(transaction_date) AS Month,
    ROUND(COUNT(transaction_id)) AS Total_Orders,
    (COUNT(transaction_id) - LAG(COUNT(transaction_id), 1)
     OVER (ORDER BY MONTH(transaction_date))) /
    LAG(COUNT(transaction_id), 1)
     OVER (ORDER BY MONTH(transaction_date)) * 100 AS mom_increase_percentage
FROM coffee_shop_sales
WHERE MONTH(transaction_date) IN (4, 5)
GROUP BY MONTH(transaction_date)
ORDER BY MONTH(transaction_date);

```

Q3 — Total Quantity Sold KPI + MoM Growth

```

-- Total Quantity Sold for May
SELECT SUM(transaction_qty) AS Total_Quantity_Sold
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5;

-- MoM Quantity Difference & Growth %
SELECT
    MONTH(transaction_date) AS Month,
    ROUND(SUM(transaction_qty)) AS Total_Quantity_Sold,
    (SUM(transaction_qty) - LAG(SUM(transaction_qty), 1)
     OVER (ORDER BY MONTH(transaction_date))) /
    LAG(SUM(transaction_qty), 1)
     OVER (ORDER BY MONTH(transaction_date)) * 100 AS mom_increase_percentage
FROM coffee_shop_sales
WHERE MONTH(transaction_date) IN (4, 5)
GROUP BY MONTH(transaction_date)
ORDER BY MONTH(transaction_date);

```

Q4 — Calendar Heat Map (Single Day KPIs)

```
-- Sales, Qty, Orders for a specific date (18 May 2023)
SELECT
    CONCAT(ROUND(SUM(unit_price * transaction_qty) / 1000, 1), 'K') AS Total_Sales,
    CONCAT(ROUND(SUM(transaction_qty) / 1000, 1), 'K') AS Total_Quantity_Sold,
    CONCAT(ROUND(COUNT(transaction_id) / 1000, 1), 'K') AS Total_Orders
FROM coffee_shop_sales
WHERE transaction_date = '2023-05-18';
```

Q5 — Weekend vs Weekday Sales

```
SELECT
    CASE
        WHEN DAYOFWEEK(transaction_date) IN (1, 7) THEN 'Weekends'
        ELSE 'Weekdays'
    END AS date_type,
    SUM(unit_price * transaction_qty) AS Total_Sales
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5
GROUP BY
    CASE
        WHEN DAYOFWEEK(transaction_date) IN (1, 7) THEN 'Weekends'
        ELSE 'Weekdays'
    END;
```

Q6 — Sales by Store Location

```
SELECT
    store_location,
    CONCAT(ROUND(SUM(unit_price * transaction_qty) / 1000, 2), 'K') AS Total_Sales
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5
GROUP BY store_location
ORDER BY SUM(unit_price * transaction_qty) DESC;
```

Q7 — Average Daily Sales + Above/Below Average Flag

```
-- Average daily sales for May
SELECT AVG(Total_Sales) AS Avg_Sales
FROM (
    SELECT SUM(unit_price * transaction_qty) AS Total_Sales
    FROM coffee_shop_sales
    WHERE MONTH(transaction_date) = 5
    GROUP BY transaction_date
) AS Internal_Query;

-- Daily sales vs average – flag each day as Above/Below Average
SELECT
    day_of_month,
    CASE
```

```

        WHEN total_sales > avg_sales THEN 'Above_Avg'
        WHEN total_sales < avg_sales THEN 'Below_Avg'
        ELSE 'Avg'
    END AS sales_status,
    total_sales
FROM (
    SELECT
        DAY(transaction_date) AS day_of_month,
        SUM(unit_price * transaction_qty) AS total_sales,
        AVG(SUM(unit_price * transaction_qty)) OVER() AS avg_sales
    FROM coffee_shop_sales
    WHERE MONTH(transaction_date) = 5
    GROUP BY DAY(transaction_date)
) AS sales_data
ORDER BY day_of_month;

```

Q8 — Sales by Product Category (Top 10)

```

SELECT
    product_category,
    SUM(unit_price * transaction_qty) AS Total_Sales
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5
GROUP BY product_category
ORDER BY SUM(unit_price * transaction_qty) DESC
LIMIT 10;

```

Q9 — Sales by Hour & Day of Week

```

-- Overall May KPIs
SELECT
    CONCAT(ROUND(SUM(transaction_qty * unit_price) / 1000, 2), 'K') AS Total_Sales,
    SUM(transaction_qty) AS Total_Qty_Sold,
    COUNT(*) AS Total_Orders
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5;

-- Hourly Sales for May
SELECT
    HOUR(transaction_time) AS Hours,
    SUM(transaction_qty * unit_price) AS Total_Sales
FROM coffee_shop_sales
WHERE MONTH(transaction_date) = 5
GROUP BY HOUR(transaction_time)
ORDER BY HOUR(transaction_time);

-- Day of Week Sales for May
SELECT
    CASE
        WHEN DAYOFWEEK(transaction_date) = 2 THEN 'Monday'
        WHEN DAYOFWEEK(transaction_date) = 3 THEN 'Tuesday'
        WHEN DAYOFWEEK(transaction_date) = 4 THEN 'Wednesday'
        WHEN DAYOFWEEK(transaction_date) = 5 THEN 'Thursday'
        WHEN DAYOFWEEK(transaction_date) = 6 THEN 'Friday'
        WHEN DAYOFWEEK(transaction_date) = 7 THEN 'Saturday'
        ELSE 'Sunday'
    END

```

```
END AS Day_of_Week,  
    ROUND(SUM(transaction_qty * unit_price)) AS Total_Sales  
FROM coffee_shop_sales  
WHERE MONTH(transaction_date) = 5  
GROUP BY Day_of_Week;
```



DAX Measures & Calculated Columns Used in Power BI

All KPI cards, MoM comparisons, and dynamic visuals in the dashboard are powered by the following DAX measures and calculated columns.



Date Table (Calculated Table)

```
Date_Table = CALENDAR(MIN(Transactions[transaction_date]), MAX(Transactions[transaction_date]))
```



Calculated Columns on Date_Table

```
-- Month Name  
Month Name = FORMAT(Date_Table[Date], "MMM YYYY")  
  
-- Month Number (for sorting)  
Month Number = MONTH(Date_Table[Date])  
  
-- Week Day Name  
Week Day = FORMAT(Date_Table[Date], "DDD")  
  
-- Week Day Number (for sorting Mon=1 to Sun=7)  
Week Day Number = WEEKDAY(Date_Table[Date], 2)  
  
-- Hour (extracted from transaction_time in Transactions table)  
Hour = HOUR(Transactions[transaction_time])
```



Core KPI Measures

```
-- Total Sales  
Total Sales = SUM(Transactions[transaction_qty] * Transactions[unit_price])  
  
-- Total Orders  
Total Orders = COUNT(Transactions[transaction_id])  
  
-- Total Quantity Sold  
Total Qty Sold = SUM(Transactions[transaction_qty])
```



Month-on-Month (MoM) Measures

```
-- Previous Month Sales  
PM Sales = CALCULATE([Total Sales], DATEADD(Date_Table[Date], -1, MONTH))
```

```
-- MoM Sales Difference
MOM Sales Diff = [Total Sales] - [PM Sales]

-- MoM Sales Growth %
MOM Sales Growth = DIVIDE([MOM Sales Diff], [PM Sales], 0)

-- Previous Month Orders
PM Orders = CALCULATE([Total Orders], DATEADD(Date_Table[Date], -1, MONTH))

-- MoM Orders Difference
MOM Orders Diff = [Total Orders] - [PM Orders]

-- MoM Orders Growth %
MOM Orders Growth = DIVIDE([MOM Orders Diff], [PM Orders], 0)

-- Previous Month Quantity
PM Qty = CALCULATE([Total Qty Sold], DATEADD(Date_Table[Date], -1, MONTH))

-- MoM Quantity Difference
MOM Qty Diff = [Total Qty Sold] - [PM Qty]

-- MoM Quantity Growth %
MOM Qty Growth = DIVIDE([MOM Qty Diff], [PM Qty], 0)
```

Sales Trend Measures

```
-- Average Daily Sales (for trend line reference)
Average Sales = AVERAGEX(VALUES(Date_Table[Date]), [Total Sales])

-- Sales Above / Below Average Flag (used in calendar colour coding)
Sales vs Avg =
VAR Avg = [Average Sales]
VAR Current = [Total Sales]
RETURN
    IF(Current > Avg, "Above Average", IF(Current < Avg, "Below Average", "Average"))
```

Dynamic KPI Card Formatting

```
-- Sales Icon (for MoM direction indicator on KPI card)
Sales Icon =
IF([MOM Sales Growth] > 0, "▲ " & FORMAT([MOM Sales Growth], "0.0%"),
    "▼ " & FORMAT(ABS([MOM Sales Growth]), "0.0%"))
```

Dashboard Screenshot

Coffee Shop Sales Dashboard

How to Use

1. Download the `.pbix` file from this repository

2. Open in **Power BI Desktop** (free download from Microsoft)
 3. Use the **Month filter** on the left panel to switch between January–June 2023
 4. Hover over the **Calendar chart** to see daily breakdowns via tooltip
 5. Hover over the **Day & Hour matrix** to see hourly performance via tooltip
 6. All visuals are cross-filtered — click any bar, slice or cell to drill into that segment
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Key Takeaways for Business

1. **Protect the morning rush** — 7–10 AM is make-or-break. Staff fully, prep inventory the night before, minimise wait times.
 2. **Barista Espresso is your anchor product** — ensure it is always available, well-priced, and prominently positioned.
 3. **All 3 stores are healthy** — no underperforming location. Focus on consistent service standards.
 4. **Weekend revenue gap is an opportunity** — targeted weekend promotions or family bundles could unlock \$50K+ in incremental annual revenue.
 5. **Upselling premium and large-size products** — top revenue products are all large-size. Training staff to upsell could push AOV above \$5.00.
 6. **Tea is your second biggest category** — don't just market as a coffee shop. Tea drinkers are 28% of revenue and deserve dedicated menu attention.
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